

DSA06 – DATA HANDLING AND VISUALIZATION

HIGH-END ASSIGNMENT

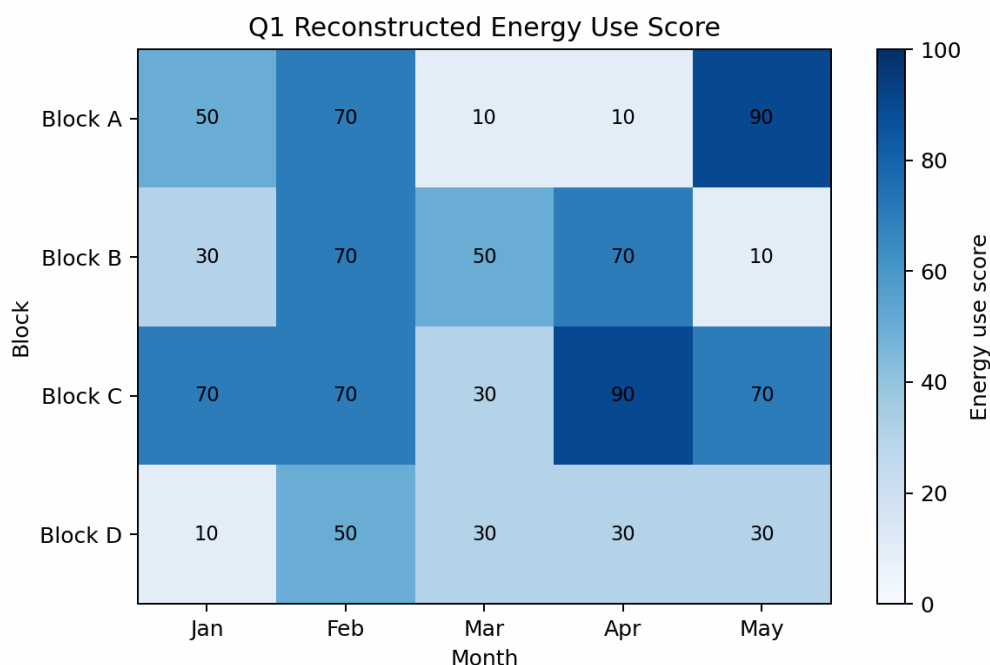
Visualizing Data • Colour Scales • Coordinates • ggplot2 • Polar Coordinates

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Reg. No.	192325068	Course	DSA06 – Data Handling and Visualization
Assignment	High-End Assignment	Coverage	Q1–Q5

Q1. Visual to Table: Reading a Colour Scale

(a) The legend contains five ordered classes: 0–20, 20–40, 40–60, 60–80 and 80–100. Reading the shade of every cell from the heat map gives the following reconstructed table.

Block / Month	Jan	Feb	Mar	Apr	May
Block A	40–60	60–80	0–20	0–20	80–100
Block B	20–40	60–80	40–60	60–80	0–20
Block C	60–80	60–80	20–40	80–100	60–80
Block D	0–20	40–60	20–40	20–40	20–40



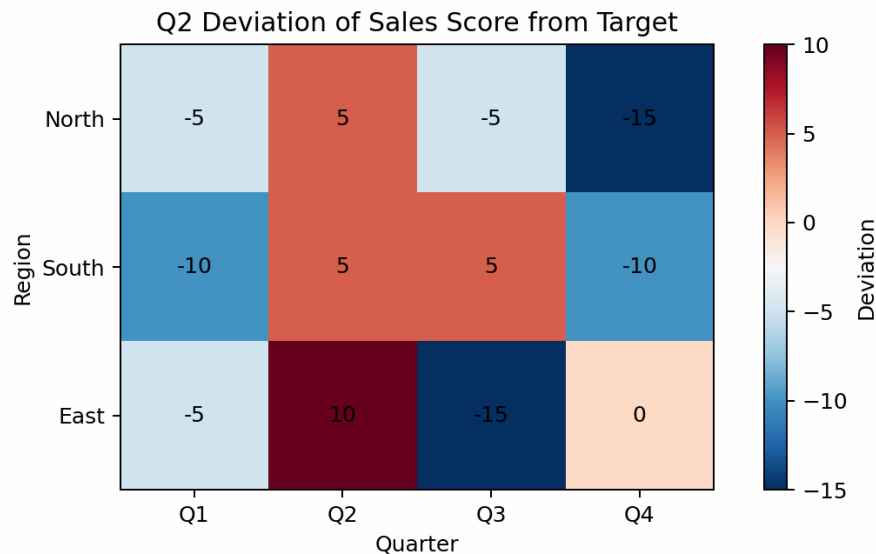
(b) The colour scale is a sequential scale. The classes progress from a very light blue for low values to a dark blue for high values. A sequential palette is appropriate when the variable has a meaningful low-to-high ordering and there is no meaningful central midpoint.

(c) The highest class is 80–100. The cells in this class are Block A–May and Block C–April. Darker colour intuitively represents a higher energy-use score because the palette increases in visual intensity as the numeric value increases.

Q2. Visual to R Code: Choosing an Effective Colour Scale

10 marks

Given data The deviation-from-target values are: North = (-5, 5, -5, -15), South = (-10, 5, 5, -10), and East = (-5, 10, -15, 0) for Q1–Q4.



(a) A diverging blue–white–red palette is appropriate because the data have a meaningful midpoint at zero. Negative deviations can be shown in blue, values close to zero in a neutral colour, and positive deviations in red.

```
library(ggplot2)

# Data in long format
df <- data.frame(
  Region = rep(c("North", "South", "East"), each = 4),
  Quarter = rep(c("Q1", "Q2", "Q3", "Q4"), times = 3),
  Deviation = c(-5, 5, -5, -15, -10, 5, 5, -10, -5, 10, -15, 0)
)

ggplot(df, aes(x = Quarter, y = Region, fill = Deviation)) +
  geom_tile() +
  scale_fill_gradient2(low = "blue", mid = "white", high = "red", midpoint = 0) +
  labs(fill = "Deviation") +
  theme_minimal()
```

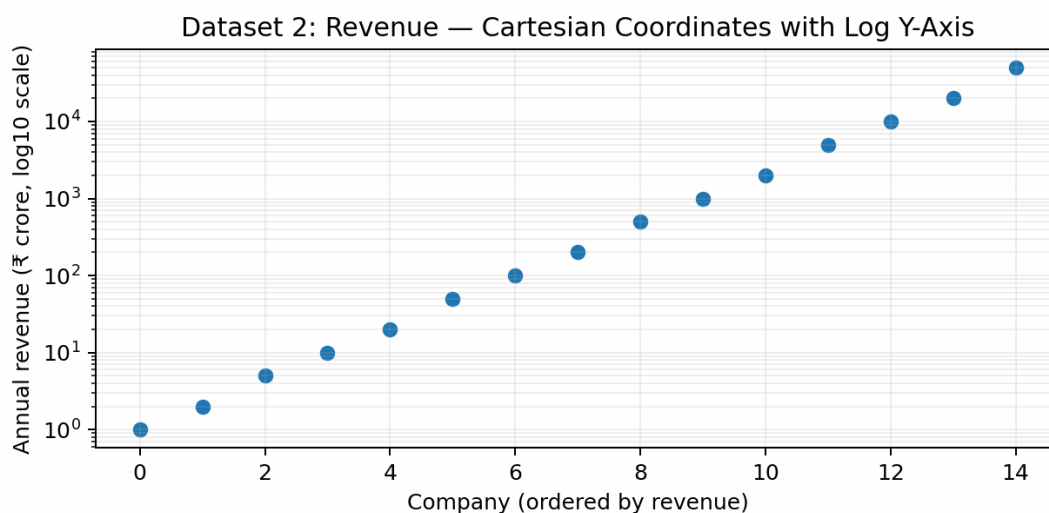
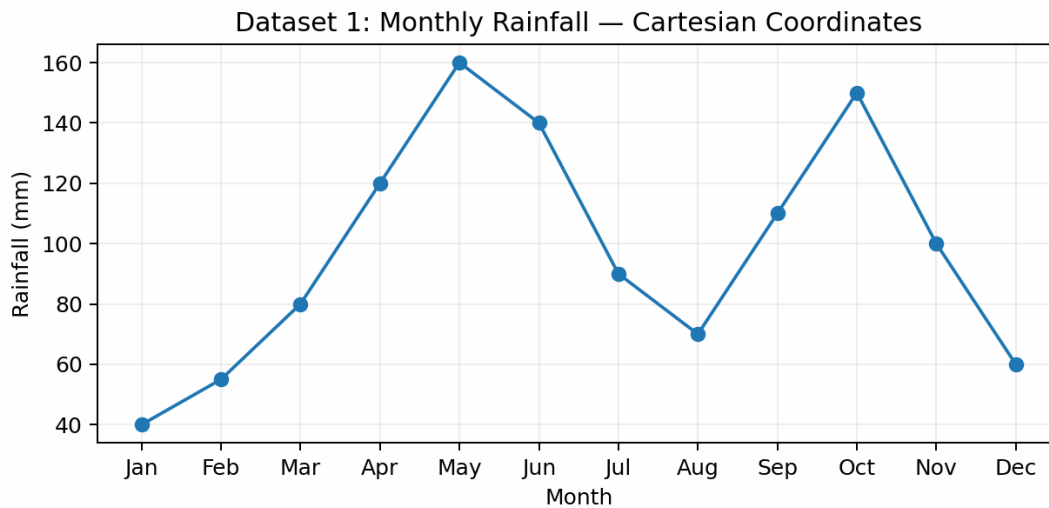
(b) A diverging scale is preferable to a sequential or rainbow scale because zero is the decision boundary: negative values mean sales are below target, while positive values mean sales are above target. A sequential scale would hide this two-sided meaning, while a rainbow palette introduces unnecessary hue changes and makes ordering harder to perceive.

Example 1: -15 is strongly below target and should appear near the blue extreme. **Example 2:** +10 is above target and should appear near the red extreme.

Q3. Design Decision: Choosing Coordinate Systems and Axes

10 marks

(a) Dataset 1 – monthly rainfall over a year: use a Cartesian coordinate system. Put month on the x-axis and rainfall on the y-axis. This makes month-to-month comparison and trends straightforward. Dataset 2 – annual revenue of 30 companies ranging from ₹1 crore to ₹50,000 crore: use Cartesian coordinates with revenue on the y-axis and a logarithmic transformation. The very wide range would otherwise compress small and medium companies near the baseline.



(b) The appropriate transformation for Dataset 2 is a base-10 logarithmic transformation of the revenue axis. A log scale changes multiplicative differences into more visually manageable equal intervals. For example, ₹10 crore and ₹100 crore differ by a factor of 10, and ₹1,000 crore and ₹10,000 crore also differ by a factor of 10; on a log axis these equal ratios occupy equal visual distances.

(c) In ggplot2, the main function calls are:

```
# Dataset 1: Cartesian coordinates
ggplot(rainfall_df, aes(x = Month, y = Rainfall)) +
  geom_line(group = 1) + geom_point()

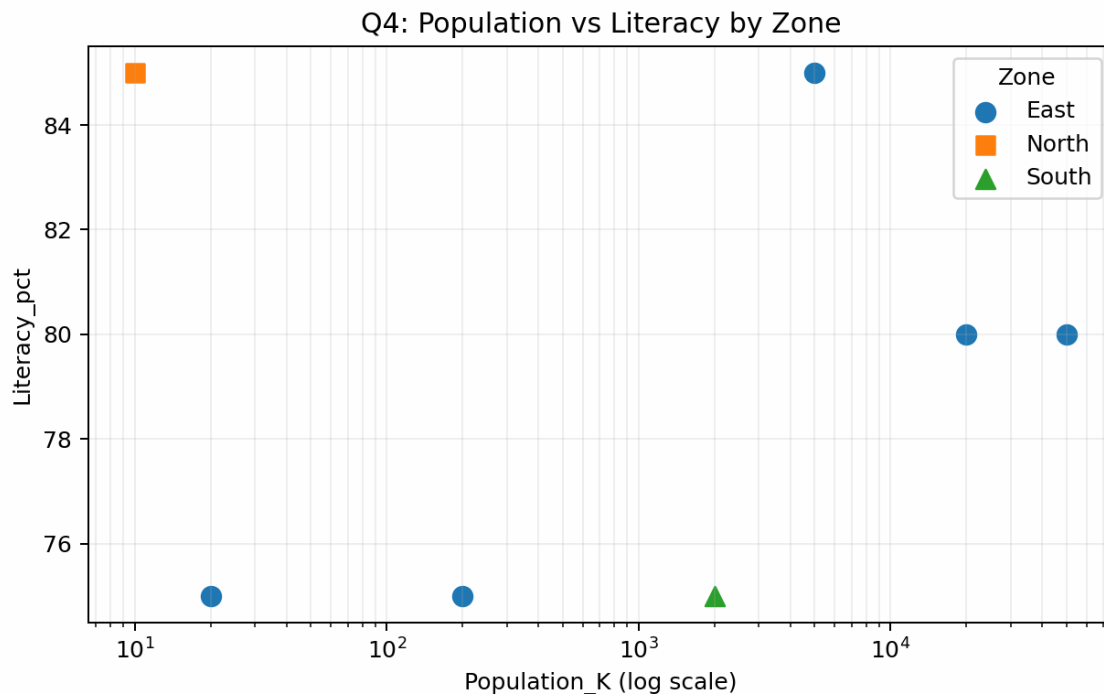
# Dataset 2: Cartesian coordinates + log10 y-axis
ggplot(revenue_df, aes(x = Company, y = Revenue)) +
  geom_point() +
  scale_y_log10()
```

Design rule: Use the coordinate system that makes the structure of the data easiest to compare. Do not use a log axis merely because it looks sophisticated; it is justified here because the revenue range spans several orders of magnitude.

Q4. R Code to Visual: Aesthetics and a Log Axis

10 marks

Data extracted from the R program Vellore (20K, 75, East), Theni (200K, 75, East), Trichy (10K, 85, North), Salem (20,000K, 80, East), Karur (5,000K, 85, East), Erode (50,000K, 80, East), and Chennai (2,000K, 75, South). The x-variable is Population_K, y is Literacy_pct, and colour represents Zone.



(a) The chart is a scatter plot with seven points. The x-axis is Population_K on a base-10 logarithmic scale; the y-axis is Literacy_pct on a normal linear scale. The points are separated by the categorical Zone variable, producing a legend with East, North and South. The approximate x positions are 10, 20, 200, 2,000, 5,000, 20,000 and 50,000, while literacy values are 75%, 80% or 85%.

(b) Aesthetics: x = Population_K, y = Literacy_pct, colour = Zone, point size = 3. Scale: x uses log10 transformation. Coordinate system: standard Cartesian coordinates; only the x-axis is transformed.

(c) A colour-blind-safe alternative is the Okabe–Itô palette. In ggplot2, one simple one-line replacement is:

```
scale_colour_manual(values = c("East"="#0072B2", "North"="#D55E00", "South"="#009E73"))
```

Why this matters: The categorical colours should remain distinguishable without depending only on red-versus-green discrimination. The log x-axis is essential because the population values vary from 10K to 50,000K.

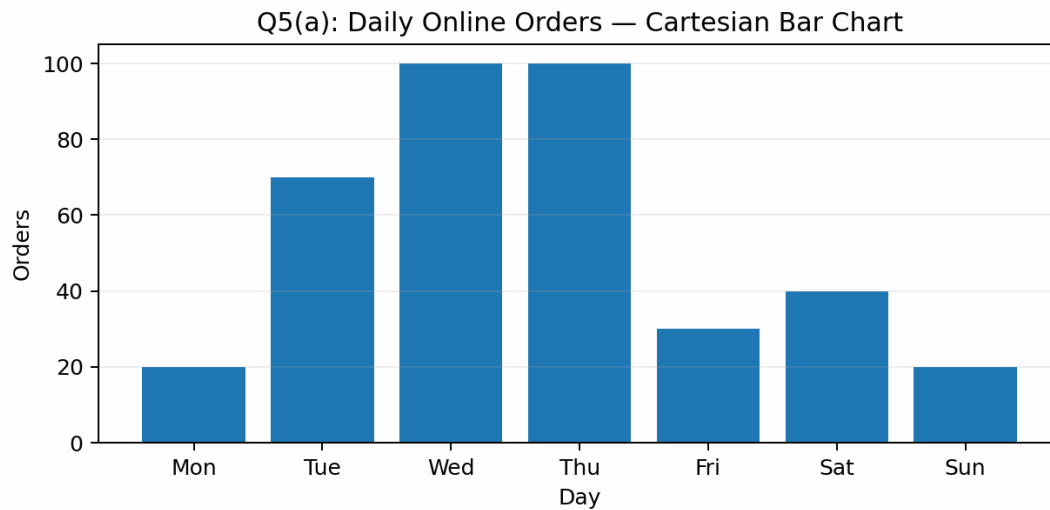
Q5. Table to Visual: Cartesian versus Polar Coordinates

10 marks

Given data Online orders across seven days: Mon 20, Tue 70, Wed 100, Thu 100, Fri 30, Sat 40, Sun 20.

Day	Orders
Mon	20
Tue	70
Wed	100
Thu	100
Fri	30
Sat	40
Sun	20

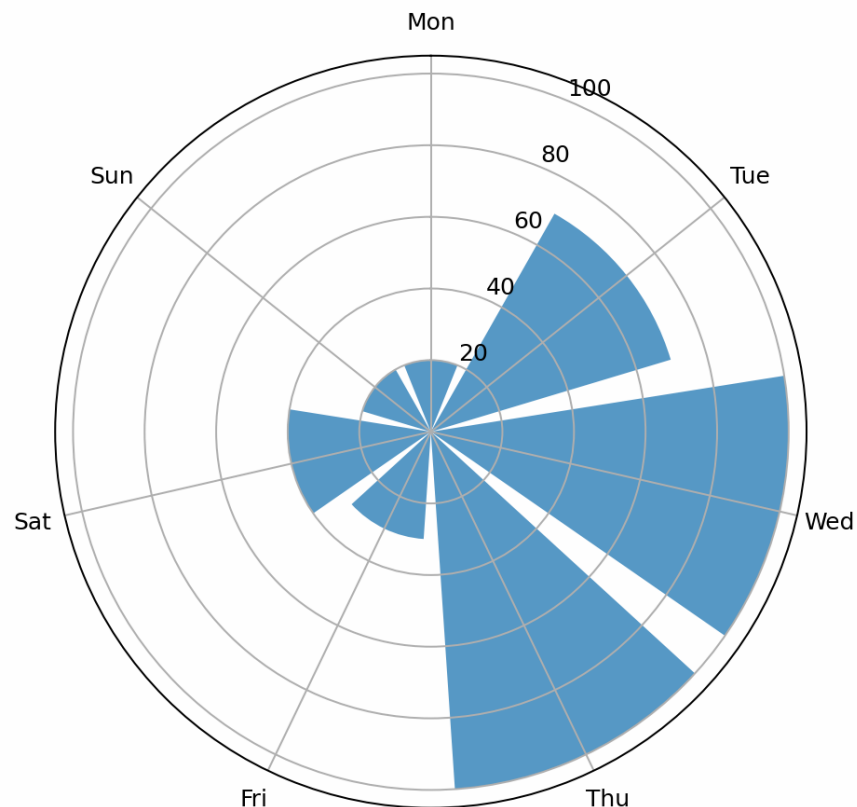
(a) In Cartesian coordinates, place the seven days on the x-axis and number of orders on the y-axis. A bar chart is the clearest choice because the variable is a discrete daily count.



(b) In polar coordinates, the seven days are placed at equal angular intervals around a circle. One full circle is 360° , so each day represents $360^\circ/7 \approx 51.43^\circ$. The radius of each bar represents the number of orders.

Day	Orders (radius)	Angle from Monday
Mon	20	0.00°
Tue	70	51.43°
Wed	100	102.86°
Thu	100	154.29°
Fri	30	205.71°
Sat	40	257.14°
Sun	20	308.57°

Q5: Online Orders — Polar Bar Chart



(c) For this dataset, Cartesian coordinates are better. The main purpose is comparing exact order counts across days, and bars on a linear y-axis make 20, 40, 70 and 100 easy to compare. Polar coordinates have an advantage when the variable is genuinely cyclic, because the end and beginning of the cycle visually meet. However, the polar version introduces angular position and radial length, which makes precise magnitude comparisons harder.

Advantage example: A weekly cycle can be recognized immediately as a circular sequence from Monday through Sunday. **Drawback example:** A 100-order bar and a 70-order bar are harder to compare accurately by radial length than by Cartesian height.